🔑 **Answer Key — Teacher Copy · Lesson 3-8**

MSTAR PRACTICE WORKSHEET

# Lesson 3-8: Solve Problems with Unit Rates

Grade 6 Mathematics · Standard 6.AT.2

*These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.*

## Part A — Skills Check

One point each.

**Question 1** SELECTED RESPONSE — CORRECT: D

Which is the better buy: 6 bouncy balls for $4.50 or 10 bouncy balls for $8.00?

A. 10 for $8.00 — $0.80 each

B. They cost the same per ball

C. Not enough information

D. 6 for $4.50 — $0.75 each

$4.50 ÷ 6 = $0.75 each. $8.00 ÷ 10 = $0.80 each. $0.75 < $0.80, so 6 for $4.50 is the better buy.

**Question 2** SELECTED RESPONSE — CORRECT: B

A faucet drips 15 ounces of water in 5 minutes. What is the unit rate?

A. 0.33 ounces per minute

B. 3 ounces per minute

C. 5 ounces per minute

D. 15 ounces per minute

15 ÷ 5 = 3 ounces per minute.

**Question 3** SELECTED RESPONSE — CORRECT: C

A printer prints 42 pages in 6 minutes. How many pages does it print in 15 minutes at the same rate?

A. 70 pages

B. 90 pages

C. 105 pages

D. 120 pages

Unit rate: 42 ÷ 6 = 7 pages per minute. In 15 minutes: 7 × 15 = 105 pages.

**Question 4** SELECTED RESPONSE — CORRECT: B

Store A sells 5 notebooks for $8. Store B sells 3 notebooks for $5.25. Which has the lower price per notebook?

A. Store B ($1.75 each)

B. Store A ($1.60 each)

C. Same price

D. Cannot determine

Store A: $8 ÷ 5 = $1.60. Store B: $5.25 ÷ 3 = $1.75. $1.60 < $1.75, so Store A is cheaper per notebook.

**Question 5** SELECTED RESPONSE — CORRECT: C

A machine dispenses 24 tokens in 3 minutes. At this rate, how many tokens does it dispense in 10 minutes?

A. 60 tokens

B. 72 tokens

C. 80 tokens

D. 240 tokens

Unit rate: 24 ÷ 3 = 8 tokens per minute. In 10 minutes: 8 × 10 = 80 tokens.

**Question 6** SELECTED RESPONSE — CORRECT: C

A factory produces 360 widgets in 8 hours. At the same rate, how many widgets are produced in a 12-hour shift?

A. 360 widgets

B. 480 widgets

C. 540 widgets

D. 720 widgets

Unit rate: 360 ÷ 8 = 45 widgets per hour. In 12 hours: 45 × 12 = 540 widgets.

## Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

**Question 7** TWO-PART QUESTION

**Part A — correct answer: C**

A store sells 8 notebooks for $20. At the same unit price, what is the cost of 14 notebooks?

A. $26.00

B. $28.00

C. $35.00

D. $40.00

Unit price = 20 ÷ 8 = $2.50 per notebook. Then 14 × 2.50 = $35.00.

- **A:** That adds $1 for each of the 6 extra notebooks. Each notebook costs $2.50, so 6 more cost $15.
- **B:** That adds the 8 notebooks to the $20. Find the price of ONE notebook first: 20 divided by 8.
- **D:** That doubles the $20, which would buy 16 notebooks. Multiply the unit price by 14.

**Part B — correct answer: C**

Which sequence of steps solves Part A?

A. Multiply the total cost by the new quantity, then divide by the old quantity's units.

B. Add the difference in notebooks to the original price.

C. Divide the total cost by the quantity to find the unit price, then multiply by the new quantity.

D. Subtract the quantities and multiply by the total cost.

Finding the unit price first gives the cost of one item, which can then be scaled to any quantity.

**Question 8** WRITTEN RESPONSE · 2 POINTS

**Type II Reasoning: Adding Instead of Scaling**

> Rosa knew 8 notebooks cost $20 and wanted the cost of 14 notebooks. She wrote: '14 is 6 more than 8, so add $6: the cost is $26.'

Explain Rosa's misconception and give the correct cost.

**Model answer:** The unit price is 20 ÷ 8 = $2.50. Six more notebooks cost 6 × $2.50 = $15, so the total is $35.00. Rosa added the DIFFERENCE IN COUNT to the price, mixing notebooks and dollars.

| | |
|---|---|
| **2 points** | Student explains that the relationship is multiplicative, not additive — 6 more notebooks cost 6 × $2.50 = $15, not $6 — and gives $35.00. |
| **1 point** | Student gives $35.00 without explaining why adding is wrong. |
| **0 points** | Student agrees with Rosa or gives an unrelated response. |

## Part C — Show Your Thinking

Model answer and scoring guidance.

**Question 9** WRITTEN RESPONSE · 2 POINTS

Two arcade supply companies offer token deals. Company X: 300 tokens for $36. Company Y: 500 tokens for $55. The arcade needs 1,000 tokens. Find the unit rate for each company, determine which is cheaper per token, and calculate the total cost of buying 1,000 tokens from the better deal.

**Model answer:** Company X charges $36 ÷ 300 = $0.12 per token and Company Y charges $55 ÷ 500 = $0.11 per token, so Y is cheaper for each token even though its package costs more. At $0.11 per token, 1,000 tokens cost 1,000 × $0.11 = $110.

*Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.*

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Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.